CONFOEDERATIO,
RESEARCH DIVISION (CRD).

DEMOGRAPHIC LITERATURE REVIEW: THE WILD FIELDS AND BUDJAK (10000BC - 1400AD).

Natural Hydrological Sub-Regions, Budjak Maritime Corridor, Kurgan Palaeodemography, Classical Chora Networks and Medieval Steppe-Riverine Symbiosis.

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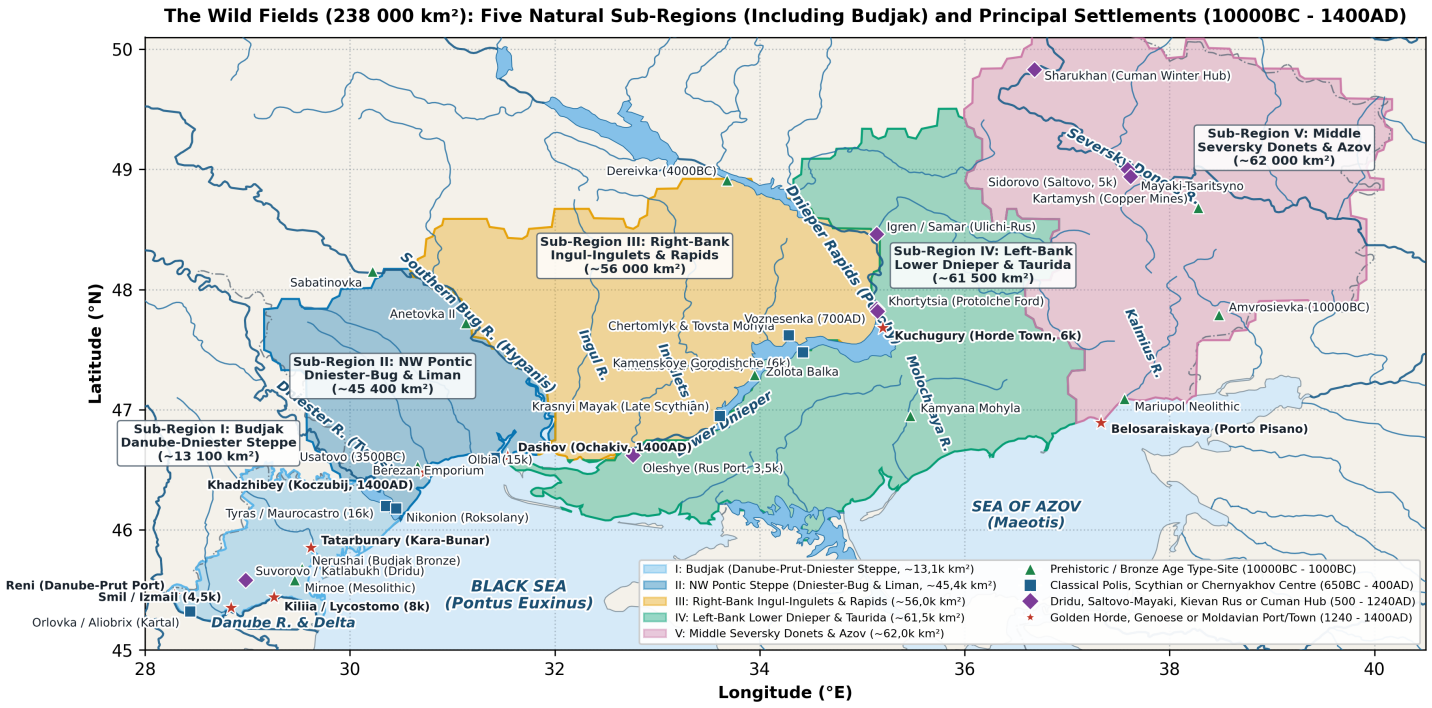


Figure 1. Cartographic overview of the Wild Fields (238 000 km²) showing the five natural hydrological sub-regions (with Budjak as Sub-Region I), steppe river corridors and principal settlements (10000BC - 1400AD).

Geographical Definition, Ecological Regimes and Five-Part Natural Sub-Regional Partition

Spatial Boundaries and Natural Steppe Hydrological Corridors

The analytical region designated as **The Wild Fields** (`wild_fields`, historical Polish *Dzikie Pola*, Latin *Loca Deserta*, Ukrainian *Dyke Pole* and the western wing of *Desht-i Kipchak*) encompasses the entire southern Ukrainian Pontic-Azov steppe zone north of the Perekop Isthmus and south of the WWF Forest-Steppe ecotone (`forest_wwf_cr.asc`), spanning approximately **238 000 km²** (4 011 5-arcminute HYDE grid cells covering 231 135 km² of land and estuarine cell area) between 45,2°N and 49,9°N latitude and 28,2°E and 40,2°E longitude [1][2]. This literature review and its companion dasymetric calibration cover the period from the Pleistocene-Holocene transition (**10000BC**) to the late Golden Horde, Genoese, Moldavian and

Lithuanian benchmark of **1400AD**; post-1400AD developments (the Ottoman-Tatar frontier, Zaporozhian *Siches* and Imperial/Modern colonisation) require separate regime handling. The Crimean Peninsula south of Perekop and Chongar (~26 945 km^2) is treated in the companion `crimea` synthesis, while the forest-steppe principalities of Galicia-Volhynia (`galicia_and_volhynia`), Kiev-Pereyaslavl (`kiev_and_pereyaslavl`) and Chernigov-Seversk (`chernigov_and_novgorod_seversk`) bound the Wild Fields along the natural forest-steppe line to the north.

Across the Holocene (10000BC - 1400AD), demographic carrying capacity in the Wild Fields was governed by the contrast between two ecological zones:

1. **Open Interfluvial Feather-Grass Steppe (*Plakor*)**: Gently undulating loess plateaus covered by *Stipa-Festuca* chernozems and southern chestnut soils receiving 350 - 480 mm of annual precipitation. Prior to modern deep well-drilling, the interfluvial watersheds lacked permanent summer surface water and functioned primarily as meridian grazing corridors for mobile pastoralists [1][3].
2. **Incised River Valleys, Rapids, Floodplain Forests (*Plavni*), Freshwater Lakes and Coastal Estuaries (*Limany*)**: The great southward-flowing rivers - the Lower Danube (Chilia/Kiliia arm) and its Budjak lakes (Yalpug, Kugurlui, Katlabukh, Kitai, Kagul), the Dniester (*Tyras*), Tiligul, Southern Bug (*Hypanis*), Ingul, Ingulets, Lower Dnieper (*Borysthenes*) with its nine granite rapids (*Porohy*) and the *Veliky Lug* ("Great Meadow") gallery forest, Samara, Orel, Konka, Molochnaya, Kalmius, Mius and Middle Seversky Donets - carved sheltered micro-ecologies rich in gallery timber, fresh water, anadromous sturgeon and cyprinid fisheries, winter reed-bed fodder and defensive bluffs [4][5][6][7].

Five-Part Natural Sub-Regional Framework (Including Budjak)

To ensure that both the dasymetric areal mask (`wild_fields_areal_mask.png`) and vector cartography (Figure 1 and Table 1) follow strict natural hydrological and ecological boundaries while treating **Budjak** as a distinct historical-demographic region, the 238 000 km^2 domain is partitioned into **five natural sub-regions**:

- **Sub-region I: Budjak (Danube-Prut-Dniester Steppe and Liman, ~13 100 km^2; 227 HYDE grid cells)**. Bounded naturally by the Lower Danube River and Chilia Delta on the south, the Prut confluence and Moldovan border on the west, the Black Sea coast (with the Sasyk, Shagany, Alibey and Burnas lagoons) on the south-east, and the Dniester River and Dniester Liman on the north-east. Unlike the interior continental steppe, Budjak combines an open interior *plakor* (drained by the Kogilnik, Chaga, Sarata and Khadzhider rivers) with two high-carrying-capacity maritime-fluvial interfaces: the Danube floodplain lakes (Kagul, Yalpug, Katlabukh, Kitai) and the western bank of the Dniester Liman. Consequently, Budjak served continuously as the primary demographic bridge between the Balkan-Danubian world (Gumelnita, Getae, Roman Lower Moesia, First/Second Bulgarian Empire *Dridu* culture, Byzantium, Genoese *Lycostomo* and Moldavia) and the Pontic steppe (Yamnaya/Nerushai, Scythians, Sarmatians, Pechenegs, Cumans and Golden Horde *Aq Qerman*) [8][9][10][11].
- **Sub-region II: North-Western Pontic Steppe (Dniester-Bug Interfluve and Tiligul-Berezan-Bug Liman, ~45 400 km^2; 475 HYDE grid cells)**. Bounded naturally on the south-west by the Dniester River and Liman, on the north by the Kodyma River and WWF Forest-Steppe ecotone, on the north-east by the Southern Bug River (*Hypanis*) and Bug Liman, and on the south by the Black Sea coast. Historically the core of the Eneolithic Usatovo complex (Usatovo, Mayaki), the Milesian Greek *poleis* and *chora* networks of **Olbia**, **Berezan**, **Nikonion** and **Koshary**, dense Chernyakhov coastal

villages (Koblevo, Kamenka-Anchekrak), Tivertsi/Ulichi forest-steppe frontier settlements and the late fourteenth-century ports of **Khadzhibey** (*Koczubij*) and **Dashov** (*Ochakiv*) [12][13][14][15].

- **Sub-region III: Right-Bank Bug-Dnieper Interfluve, Ingul-Ingulets Uplands and Dnieper Rapids Right Bank (~56 000 km²; 922 HYDE grid cells)**. Bounded naturally on the west by the Southern Bug River, on the north by the WWF Forest-Steppe ecotone along the Dnieper Upland, and on the east and south-east by the Dnieper River (*Borysthenes*) and Dnieper-Bug Estuary. Encompassing the Syniukha, Ingul, Ingulets, Visun, Saksahan and Bazavluk basins, this sub-region hosted the Dereivka Eneolithic enclosure, the Sabatinovka and Ingul Catacomb Bronze Age heartland, the Royal Scythian giant kurgans (Chertomlyk, Tovsta Mohyla, Alexandropol) and the chain of Late Scythian right-bank stone fortresses (Zolota Balka, Gavrilovka, Krasny Mayak, Konsulovskoye, Nikolaevka-Kazatskoye) [5][16][17].
- **Sub-region IV: Left-Bank Lower Dnieper, Veliky Lug, Samara-Orel and Northern Taurida Steppe (~61 500 km²; 1 107 HYDE grid cells)**. Bounded on the west by the Dnieper River channel, on the north by the Orel (*Oril*) River and WWF Forest-Steppe ecotone, on the east by the Dnieper-Donets / Azov Upland watershed (Berda River divide), and on the south by the Black Sea, Perekop-Sivash and Sea of Azov. It contained the Vasylivka Mesolithic cemeteries, the Kamyana Mohyla sanctuary, the Mikhailovka Yamnaya stone citadel, the 1 200 ha Scythian city of Kamenskoye Gorodishche, the Voznesenka princely camp, the Kievan Rus ports of **Oleshye**, **Protolche** (Khortytsia) and **Igren-Samar**, and the Golden Horde cities of **Velyki Kuchugury**, **Mamai-Surka** and **Konski Vody** [18][19][20][21].
- **Sub-region V: Middle Seversky Donets, Donets Ridge and Kalmius-Mius Azov Basin (~62 000 km²; 1 280 HYDE grid cells)**. Bounded on the west by the Dnieper-Donets / Azov Upland watershed, on the north-west by the WWF Forest-Steppe ecotone, on the east by the Aidar-Derkul-Donets-Mius basin frontier, and on the south by the Sea of Azov. Spanning the chalk bluffs of the Middle Seversky Donets and Oskol valleys, the elevated Donets Ridge (*Donetskiy Kryazh*) and the Kalmius-Mius coastal slope, this eastern sector supported the Amvrosievka Palaeolithic camp, the Mariupol Neolithic cemetery, the Bronze Age copper mines of Kartamysh, the eighth- to tenth-century Khazar-Alanic-Bulgar urban network of the *Saltovo-Mayaki* culture (Sidorovo, Mayaki-Tsaritsyno, Sukhaya Gomolsha, Mokhnach), the Polovtsian winter capitals of **Sharukhan**, **Sugrov** and **Balin**, and the Golden Horde / Italian Azov port at **Belosaraiskaya** (*Porto Pisano*) [6][22][23].

Sub-Region	Area (% Share)	Natural Hydrological & Ecological Boundaries	Representative Pre-1400AD Anchor Centres
I. Budjak (Danube-Prut-Dniester Steppe & Liman)	13 100 (5,5%)	Danube R. & Delta (S), Prut/Moldova (W), Black Sea (SE), Dniester R. & Liman (NE)	Mirnoe, Nerushai, Orlovka (Aliobrix), Tyras / Aq Qerman (Maurocastro), Kiliia (Lycostomo), Smil (Izmail), Reni
II. NW Pontic Steppe (Dniester-Bug Interfluve & Liman)	45 400 (19,1%)	Dniester R. (SW), Kodyma & Forest-Steppe (N), Southern Bug R. & Liman (NE)	Usatovo, Mayaki, Nikonion, Berezan, Olbia, Koshary, Sabatinovka, Anetovka II, Khadzhibey, Dashov (Ochakiv)
III. Right-Bank Bug-Dnieper, Ingul-Ingulets & Rapids	56 000 (23,5%)	Southern Bug R. (W), Forest-Steppe Ecotone (N), Dnieper R. & Estuary (E/SE)	Dereivka, Chertomlyk, Tovsta Mohyla, Zolota Balka, Gavrilovka, Krasny Mayak, Bashmachka, Voznesensk

(Continued from previous page)

Sub-Region	Area (% Share)	Natural Hydrological & Ecological Boundaries	Representative Pre-1400AD Anchor Centres
IV. Left-Bank Lower Dnieper, Veliky Lug & N. Taurida	61 500 (25,8%)	Dnieper R. (W), Orel R. & Forest-Steppe (N), Azov Upland Divide (E), Perekop/Azov (S)	Vasylivka, Mikhailovka, Kamyana Mohyla, Kamenskoye, Voznesenka, Khortytsia (Protolche), Oleshye, Kuchugury
V. Middle Seversky Donets, Donets Ridge & Azov Basin	62 000 (26,1%)	Dnieper-Donets Divide (W), Forest-Steppe (NW), Aidar-Mius Frontier (E), Azov Sea (S)	Amvrosievka, Mariupol Neolithic, Kartamysh, Sidorovo, Mayaki-Tsaritsyno, Sukhaya Gomolsha, Sharukhan, Sugrov
Total Wild Fields (wild_fields, 10000BC - 1400AD)	238 000 (100,0%)	Five Natural Hydrological & Ecological Sub-Regions (4 011 Grid Cells)	191 Georeferenced Pre-1400AD Settlements in Stadester Gazetteer

Table 1. Natural Hydrological and Ecological Partition of the Wild Fields (wild_fields, 238 000 km2) into Five Analytical Sub-Regions (10000BC - 1400AD).

Prehistoric Foragers, Eneolithic Horse-Keepers and Bronze Age Agro-Pastoralists (10000BC - 1000BC)

Epigravettian, Mesolithic and Neolithic Fluvial Refugia (10000BC - 4500BC)

At the Pleistocene-Holocene transition (10000BC), the retreat of periglacial tundra-steppe and the drowning of the exposed north-western Black Sea shelf into brackish *liman* estuaries concentrated Upper Palaeolithic Epigravettian and Early Mesolithic bands along limestone canyon gorges, the Budjak coastal rivers and granite river rapids [4][14]. Archaeological surveys synthesised by Stanko and Olenkovskiy across the Southern Bug (**Anetovka II**, 47,72 N, 31,13 E in Sub-region II), the Krynka-Mius basin (**Amvrosievka** bison kill site and bone bed, 47,79 N, 38,48 E in Sub-region V) and the Lower Dnieper document specialised communal hunting of steppe bison (*Bison priscus*), wild horse (*Equus ferus*) and saiga antelope [4]. During the Kukrek and Grebeniki Mesolithic horizons (8000BC), large base camps such as **Mirnoe** (45,58 N, 29,46 E in **Sub-region I: Budjak**), the sandstone rock-shelter sanctuary of **Kamyana Mohyla** (46,95 N, 35,47 E in Sub-region IV) on the Molochnaya River and the communal cemeteries along the Dnieper Rapids (**Vasylivka I-III**, **Voloske**, **Igren VIII**, containing 30 – 50+ burials per necropolis) demonstrate incipient territoriality around high-yield fishing weirs. At 10000BC, total regional population across 238 000 km^2 is estimated at **3,8k** (0,30k in Budjak [I], 0,80k in NW Pontic [II], 0,95k in Right-Bank [III], 0,95k in Left-Bank [IV] and 0,80k in Middle Donets [V]), rising to **6,8k** by 8000BC.

During the Neolithic Atlantic climatic optimum (6000BC - 5000BC), four ceramic-using traditions established riverine and estuarine villages across the five sub-regions:

1. **Danube-Lakes and Lower Dniester Horizon in Budjak (Sub-region I, 6000 - 4500BC)**: Along the shores of Lakes Yalpug, Kagul and Katlabukh, early agro-foraging communities interacted with Balkan Starcevo-Cris and Linearbandkeramik groups, lifting Budjak's population from **1,1k** at 6000BC to **1,9k** at 5000BC [9][14].
2. **Bug-Dniester Culture (Sub-regions II & III, 6200 - 5000BC)**: Concentrated along the Southern Bug rapids between Savran, Migeya and Bazkov Ostrov (48,03 N, 30,96 E), combining impressed pottery,

flint sickle blades and early domesticates [14].

3. **Sursko-Dnieper and Dnieper-Donets II Complexes (Sub-regions III & IV, 6000 - 4500BC)**: Anchored on the islands and banks of the Dnieper Rapids (**Surskoy Island**, **Nikolskoye**, **Vovnigi** with 130+ skeletons) and the Samara-Orel confluence [24].
4. **Lower Don-Azov / Mariupol Horizon (Sub-region V, 5500 - 4500BC)**: Exemplified by the 122-burial red-ochre corridor cemetery at **Mariupol** (47,09 N, 37,56 E) at the mouth of the Kalmius River. Total Wild Fields population doubled from **14,0k** at 6000BC to **24,0k** at 5000BC.

Bolgrad-Aldeni (Gumelnita), Sredny Stog and Usatovo Maritime Chiefdoms (4500BC - 3000BC)

Between 4500BC and 3000BC, the spread of Balkan-Carpathian copper metallurgy and steppe horse management reshaped the region, with **Budjak (Sub-region I)** and the **Dniester-Bug corridor (Sub-region II)** acting as the primary demographic engines [9][25][26]:

- **The Bolgrad-Aldeni Variant of the Gumelnita Culture and the Budjak Kurgan Phenomenon (Sub-region I, 4500 - 3000BC)**: As excavated by Subbotin around Lakes Yalpug, Kagul, Katlabukh and Kitai (**Bolhrad**, **Ozernoye**, **Etulia VI**, **Plavni**, **Nagornoye**, **Suvorovo**, **Chervonyi Yar** and **Orlovka/Kartal**), sedentary Balkan Chalcolithic farmers established dense tell-like villages of 20 – 40 wattle-and-daub houses in south-western Budjak (4200 – 3800BC, **3,8k** in Sub-region I at 4000BC) [9][27]. In the mid-fourth millennium (3500 – 3000BC), interaction between late Cucuteni-Trypillia CII, Cernavoda I and the **Budjak Culture** (*Budzhakskaya kultura*, represented by thousands of Yamnaya/Usatovo kurgan burials at **Nerushai**, **Semenovka**, **Mayaki**, **Trapovka**, **Borisovka**, **Sarata** and **Tarutyne**) raised Budjak's population to **9,5k at 3500BC** and **8,5k at 3000BC** [1][9].
- **The Usatovo Complex (Sub-region II, 3500 - 3000BC)**: East of the Dniester Liman around the Khadjibey and Kuialnyk estuaries, fortified limestone-built centres at **Usatovo** (46,53 N, 30,66 E) and **Mayaki** (46,41 N, 30,27 E) presided over **14,5k** inhabitants in Sub-region II at 3500BC [14]. Combined with Budjak (9,5k), the western maritime steppes (Sub-regions I + II) held **24,0k** inhabitants (35,3% of the Wild Fields total).
- **Sredny Stog, Dereivka and Mikhailovka (Sub-regions III, IV & V, 4000 - 3000BC)**: Along the Dnieper Rapids and Lower Dnieper, horse-keeping communities at **Dereivka** (48,91 N, 33,68 E, Sub-region III) and the three-phase stone-walled Yamnaya stronghold at **Mikhailovka** (47,29 N, 33,95 E, Sub-region IV) anchored expanding wagon-pastoral lineages across the Ingulets, Molochnaya and Seversky Donets, lifting total regional population from **37,0k** at 4000BC to **68,0k** at 3500BC and **75,0k** at 3000BC [24][26].

Catacomb, Babino and the Sabatinovka-Srubnaya Sedentary Boom (3000BC - 1000BC)

During the Middle Bronze Age **Catacomb** (*Ingul* and *Donets* variants, 2800 - 2000BC) and **Babino** (*Mnogovalikovaya*, 2200 - 1800BC) horizons, pastoral density rose steadily to **95,0k** at 2000BC (8,0k in Budjak [I], 20,0k in NW Pontic [II], 24,5k in Right-Bank [III], 24,0k in Left-Bank [IV] and 18,5k in Middle Donets [V]) [1][9].

The prehistoric Bronze Age culminated during the Late Bronze Age **Sabatinovka-Noua** (1600 - 1200BC, Sub-regions I-IV), **Srubnaya** (1600 - 1100BC, Sub-regions IV-V) and **Belozerka** (1200 - 950BC) horizons [16][28][29]:

- **Budjak and the Lower Dniester-Bug (Sub-regions I & II)**: In **Budjak (Sub-region I)**, over 200 Late Bronze Age **Sabatinovka-Coslogeni-Noua** and **Belozerka (Tudora-type)** settlements lined the

Kogilnik, Sarata, Alkaliya and Danube lake shores (**Zhovtneve/Karakurt**, **Ogorodnoye**, **Suvorovo**, **Tatarbunary**, **Diviziya**, **Tuzly**, **Shabo** and **Tudora**), supporting **15,0k** inhabitants at 1000BC [9][29]. In **Sub-region II** (Dniester-Bug Interfluve, **33,0k** at 1000BC), multi-roomed limestone-foundation villages such as **Sabatinovka** (48,15 N, 30,22 E), **Voronovka II** and **Tashlyk** practiced intensive cattle-and-millet mixed farming [16].

- **Right-Bank Ingul-Ingulets (Sub-region III, 36,0k), Left-Bank Lower Dnieper (Sub-region IV, 34,0k) and Donets-Azov Copper Province (Sub-region V, 27,0k):** Across the Ingul, Ingulets and Lower Dnieper (**Zmiivka**, **Novokievka**, **Staryi Krym Ingul**), hundreds of stone-built villages flourished alongside the Srubnaya copper-mining and bronze-casting complexes of the Bakhmut basin (**Kartamysh**, **Usovo Ozero**, **Kapitanovo** in Sub-region V), bringing total Wild Fields population to **145,0k at 1000BC** [3][28].

Classical Greek Poleis, Royal Scythia, Late Scythian Fortresses and the Chernyakhov Horizon (700BC - 400AD)

Greek Poleis and Chora Networks in Budjak (Sub-Region I) and the Dniester-Bug Liman (Sub-Region II)

Beginning in the mid-seventh century BC (~647BC), Milesian Greek colonisation established dense urban and rural *chora* networks across both **Sub-region I (Budjak)** and **Sub-region II (Dniester-Bug Interfluve)** [8][12][30]:

1. **Sub-region I (Budjak: Tyras, Orlovka/Aliobrix and the Lower Danube-Dniester Chora; 27,0k at 500BC, 18,0k at 0AD, 29,0k at 300AD):**
    - On the rocky western promontory of the Dniester Liman (modern Bilhorod-Dnistrovskyi), Milesian colonists founded **Tyras** (~600BC), which grew into a fortified, coin-minting *polis* of **6,0k** inhabitants at 500BC - 300BC, surrounded by a dense western Dniester Liman *chora* (**Shabo**, **Zatoka**, **Mologa II**, **Pivdenne I**, **Belenkoye**) [8][31].
    - At the southwestern corner of Budjak overlooking the Danube crossing at the Kartal narrows (near Reni), the fortified Geto-Scythian and later Roman-era emporium of **Orlovka / Kartal** (ancient **Aliobrix**, 45,32 N, 28,44 E, accompanied by **Novoselskoye** and **Reni**) controlled riverine trade between the Danube Delta, the Prut and the interior Budjak steppe (`1,5k – 2,2k` inhabitants, where gold staters of Cyzicus and later Roman legionary bricks of *Legio V Macedonica* and *Classis Flavia Moesica* are attested) [8][32].
2. **Sub-region II (Dniester-Bug Interfluve: Berezan, Pontic Olbia, Nikonion and Koshary; 58,0k at 500BC, 40,0k at 0AD, 59,0k at 300AD):**
    - On the Bug-Dnieper Liman, **Berezan** (~647BC, 2,5k) and **Pontic Olbia** (~600BC, 50 - 55 ha walled upper and lower city with **12,0k - 15,0k** urban residents) anchored **over 150 rural *chora* villages** (*Shirokaya Balka*, *Kozyrka*, *Chertovatoe VII*, *Viktorovka I*, *Staraya Bogdanovka II*, *Didova Khata*) housing 20,0k - 25,0k Greco-Scythian grain cultivators [12][13][33].
    - On the eastern bank of the Dniester Liman opposite Tyras, **Nikonion** (Roksolany, **3,0k** at 500 - 400BC) and its *chora* (**Nadlimanskoye**) linked the Dniester trade to **Koshary** (`Odessos`, 1,4k at the Tiligul mouth) and the Odessa Bay settlements (**Zhevakhova Gora**, **Luzanovka**) [15][34].

Royal Scythia, Kamenskoye Gorodishche, Late Scythian Fortresses and the Chernyakhov Peak (500BC - 400AD)

In the interior steppes of **Sub-regions III, IV and V**, fifth- and fourth-century BC **Royal Scythia** reached its demographic zenith along the Lower Dnieper Rapids and *Veliky Lug* (*Gerrhus*) [17][18]:

- **Kamenskoye Gorodishche and the Royal Kurgans (Sub-regions III & IV)**: At **Kamenskoye Gorodishche** (47,48 N, 34,42 E in Sub-region IV, enclosing 1 200 ha with its 32 ha **Znamenka** acropolis and the 800-burial **Mamai-Hora** necropolis), a resident metallurgical and administrative population of **6,0k** (plus `3,5k – 4,2k` at Znamenka) operated alongside the **Royal Scythian Giant Kurgans** of Sub-region III (**Chertomlyk**, **Tovsta Mohyla**, **Alexandropol**) and Sub-region IV (**Solokha**, **Melitopol**, **Gaimanova Mohyla**) [17][18]. In Sub-region V (Middle Seversky Donets and Azov, **36,0k** at 500BC), Scythian forest-steppe hillforts (**Velikaya Gomolsha**, **Koropovye Khutora**) and Azov coastal camps linked the Donets corridor to the Royal Scythians. Total Wild Fields population reached a classical peak of **235,0k at 500BC** (`27,0k` in Budjak [I], `58,0k` in NW Pontic [II], `56,0k` in Right-Bank [III], `58,0k` in Left-Bank [IV] and `36,0k` in Middle Donets [V]).
- **Sarmatian Transition and the 16 Late Scythian Stone Fortresses (250BC - 200AD)**: Following the third-century BC Sarmatian advance (which lowered regional population to **165,0k at 0AD**), sedentary agriculture rebounded along both banks of the Lower Dnieper gorge (Sub-regions III and IV) via **16 bastioned stone fortresses** (**Zolota Balka**, **Krasny Mayak**, **Gavrilovka**, **Konsulovskoye**, **Nikolaevka-Kazatskoye**, **Lvove**, **Sabinka** and **Hannivka** in Sub-region III; **Lyubimovka**, **Znamenskoye**, **Gornostaevka** and **Kairy** in Sub-region IV) [5][35][36][37]. Simultaneously, Roman vexillations stationed at **Tyras** and **Orlovka/Aliobrix** (Sub-region I) and **Olbia** (Sub-region II) stabilised the coastal zone, lifting regional population to **205,0k at 200AD** [38].
- **The Chernyakhov Sedentary Peak (300AD: 240,0k) and Post-Hunnic Trough (400AD: 150,0k)**: Under the multi-ethnic **Chernyakhov Culture** (`250 – 375AD`), hundreds of stone-built and timber villages covered **Budjak** (`29,0k` in Sub-region I at **Bolhrad-Yalpug**, **Etulia**, **Chervonyi Yar**, **Kubei**, **Nagornoye**, **Belenkoye** and **Mologa II**), the **Dniester-Bug Liman** (`59,0k` in Sub-region II at **Koblevo**, **Kamenka-Anchekrak** and **Ranconia**), the **Right-Bank Ingul-Ingulets & Rapids** (`62,0k` in Sub-region III at **Bashmachka** and **Snihurivka**), the **Left-Bank Lower Dnieper** (`54,0k` in Sub-region IV) and the **Donets corridor** (`36,0k` in Sub-region V), reaching an antique high-water mark of **240,0k at 300AD** before the **Hunnic Invasion of 375AD** reduced the Wild Fields to **150,0k at 400AD** (`18,0k` in I, `36,0k` in II, `38,0k` in III, `35,0k` in IV, `23,0k` in V) [32][39][40].

Early Medieval Nomads, the Balkan-Danubian Dridu Culture in Budjak, the Saltovo-Mayaki Boom and the Cuman-Rus Condominium (400AD - 1240AD)

Antae/Penkovka Rapids Refugia, Voznesenka and the Early Medieval Trough (400AD - 750AD)

Between 400AD and 700AD, as Kutrigur/Utigur Bulgars, Avars and Old Great Bulgaria under Khan Kubrat traversed the open steppes, regional population bottomed out at **125,0k in 500AD** (`14,0k` in Budjak [I], `28,0k` in II, `34,0k` in III, `29,0k` in IV, `20,0k` in V) before recovering to **135,0k in 600AD** and **160,0k in 700AD** [40][41]. Sedentary life persisted along the Lower Danube and Dniester crossings of Budjak (**Orlovka/Kartal**, **Novoselskoye**) and the granite islands of the Dnieper Rapids (**Voloshskoye**, **Igren-Samar**, **Balka Kantsyrka** and the c. 700AD princely memorial camp at **Voznesenka** in Sub-region IV) [41].

The Balkan-Danubian (Dridu) Boom in Budjak (Sub-Region I) and the Saltovo-Mayaki Urban Surge on the Donets (Sub-Region V) (750AD - 965AD)

During the eighth through tenth centuries (750 – 965AD), the two opposite wings of the Wild Fields - **Sub-region I (Budjak)** in the south-west and **Sub-region V (Middle Seversky Donets)** in the north-east - experienced parallel sedentary demographic booms:

1. **The First Bulgarian Empire and Balkan-Danubian (*Dridu* / *Pliska-Preslav*) Settlement Network in Budjak (Sub-region I: 21,0k at 800AD, 25,0k at 900AD):** Following Asparukh's crossing of the *Onglos* in southern Budjak (680AD) and the consolidation of the First Bulgarian Empire north of the Danube, archaeological surveys around Lakes Yalpug, Kugurlui, Katlabukh, Kitai and Kagul have documented **over 70 eighth- to tenth-century sedentary Balkan-Danubian (*Dridu*) villages** (**Suvorovo I-VI**, **Bogatoye**, **Ogorodnoye**, **Kholmskoye**, **Chervonyi Yar**, **Plavni**, **Etulia**, **Nagornoye**, **Safyany/Izmail** and **Orlovka/Kartal**), protected to the north by the monumental earthen **Lower Trajan's Wall** (*Nizhniy Trayanov Val*, stretching 126 km from the Prut at Vadul-lui-Isac across lakes Kagul, Yalpug and Katlabukh to Lake Sasyk) [9][10][42]. Inhabited by mixed Bulgar-Slavic-Vlach agriculturalists and fishermen, Budjak sustained **25,0k** inhabitants at 900AD (~1,91 persons/km^2, the highest density of any sub-region in the Wild Fields).
2. **Ulichi and Tivertsi Towns in the Dniester-Bug and Dnieper-Rapid Corridors (Sub-regions II, III & IV: 47,0k, 52,0k and 44,0k at 900AD):** Along the Lower Dniester, Southern Bug and Dnieper Rapids, East Slavic **Tivertsi** and **Ulichi** settlements (**Belenkoye**, **Palanca**, **Peresechen Steppe**, **Chern**, **Voznesensk**, **Igren-Samar**) traded down the rivers toward the Black Sea [19][42].
3. **The Saltovo-Mayaki Urban Network on the Middle Seversky Donets (Sub-region V: 62,0k at 800 - 900AD):** Under the *Pax Chazarica*, Alanic and Proto-Bulgar settlers constructed **over 250 fortified and open settlements** along the Seversky Donets and Oskol valleys, led by the **120 ha Sidorovo Complex** (49,00 N, 37,59 E, **5,0k** inhabitants), **Mayaki-Tsaritsyno** (3,0k), **Sukhaya Gomolsha** (2,6k), **Mokhnach** (2,5k), **Netailovka** (2,2k), **Teplinskoye** (2,0k), **Zlivki** (1,4k) and **Krivaya Luka** (1,3k) [6][23][43][44][45]. Across all five sub-regions, Wild Fields population climbed to **215,0k in 800AD** and **230,0k in 900AD**.

Pechenegs, Kievan Rus Ports (Oleshye, Protolche, Maurocastro) and the Cuman Apogee (965AD - 1240AD)

Pecheneg migrations and Prince Sviatoslav's 965AD destruction of Khazaria burned the Donets *Saltovo* forts and disrupted open *Dridu* villages in Budjak, lowering regional population to **195,0k at 1000AD** (22,0k in Budjak [I], 40,0k in II, 46,0k in III, 42,0k in IV, 45,0k in V) [6][46]. During the eleventh and twelfth centuries (1055 – 1223AD), however, the commercial condominium of **Kievan Rus**, **Byzantium** and the **Polovtsian / Cuman** (*Kipchak*) confederation lifted regional population to **235,0k in 1100AD** and an all-time pre-modern peak of **265,0k in 1200AD** (28,0k in Budjak [I], 54,0k in NW Pontic [II], 62,0k in Right-Bank [III], 56,0k in Left-Bank [IV], 65,0k in Middle Donets [V]) [19][46][47]:

- **Budjak Maritime-Danubian Ports and the Berladniki / Brodniki Frontier (Sub-region I: 28,0k at 1200AD):** At the Danube-Dniester mouths, Byzantine, Galician-Rus (*Berladniki*), Vlach and Cuman trade revived **Maurocastro / Belgorod** (at ancient Tyras, 3,0k – 3,5k), **Kiliia** (2,5k), **Smil / Izmail** (1,8k), **Reni** (1,0k) and **Orlovka/Kartal** (1,2k) [10][11].
- **Kievan Rus Dnieper Outposts (Sub-region IV):** At the head of the Dnieper estuary, the Kievan Rus sea-port of **Oleshye** (46,62 N, 32,76 E, **3,5k** at 1200AD) served as the maritime staple for Kiev's trade with Constantinople, paired upstream with the Khortytsia horse-ferry and *Brodnik* staging town of

Protolche (47,79 N, 35,15 E, **2,2k**) and the 25 ha Samara confluence town of **Igren-Samar** (**2,8k**) [19][48].

- **Polovtsian Winter Capitals and *Kamennaya Baba* Sanctuaries (Sub-regions III, IV & V)**: Along the Middle Seversky Donets (Sub-region V), semi-sedentary Cuman-Alanic winter towns flourished at **Sharukhan** (49,83 N, 36,68 E, **4,5k**), **Sugrov** (**3,2k**) and **Balin** (**2,8k**), while over 2 500 Polovtsian stone statues (*kamennye baby*) and elite kurgans (such as the **Chingul Kurgan** on the Molochnaya in Sub-region IV) demarcated twelfth- and early thirteenth-century clan pastures [22][46].

The Ulus of Jochi (Golden Horde), Genoese-Moldavian Budjak Ports, Lithuanian Castles and the 1400AD Cutoff (1240AD - 1400AD)

Mongol Conquest and the Golden Horde / Genoese Urban Revival in Budjak and Veliky Lug (1240AD - 1345AD)

The Mongol campaigns of 1223AD (Battle of the Kalka River in Sub-region V) and 1238 - 1240AD contracted the population of the Wild Fields from **245,0k in 1240AD** to **165,0k in 1250AD** (`18,0k` in Budjak [I], `34,0k` in II, `38,0k` in III, `36,0k` in IV, `39,0k` in V) [47][48]. Under the western *Ulus of Jochi* (Prince Nogai `d. 1299AD`, Khan Toqta `1291 – 1312AD` and Khan Uzbek `1313 – 1341AD`), however, the region achieved a major commercial and urban resurgence to **210,0k at 1300AD** [20][21][49]:

1. **Budjak's Golden Horde and Genoese Port Metropolis (Sub-region I: 28,0k at 1300AD)**: Under Nogai and Toqta, **Budjak** became the premier western maritime hub of the Golden Horde. At the Dniester mouth, **Aq Qerman / Maurocastro / Cetatea Alba** (46,20 N, 30,35 E) minted its own coinage, hosted Genoese, Armenian, Greek, Jewish, Moldavian and Tatar merchant quarters, and reached **15,0k - 16,0k** inhabitants at its `1300 – 1345AD` peak [10][11]. On the Chilia arm of the Danube Delta, the Genoese-notarial port of **Kiliia / Lycostomo** (45,44 N, 29,26 E, where Antonio di Ponzò's `1360 – 1361AD` notarial registers document massive grain, wax, salt and wine exports to Pera and Genoa) housed **6,5k - 8,0k** inhabitants, flanked by **Smil / Izmail** (`3,5k – 4,5k`), **Reni** (`1,3k`) and the interior Golden Horde camp of **Kara-Bunar / Tatarbunary** (`1,1k`) [11].
2. **Velyki Kuchugury, Mamai-Surka and Konski Vody in the Left-Bank Lower Dnieper (Sub-region IV: 48,0k at 1300AD)**: In the *Veliky Lug* floodplain, Uzbek and Jani Beg built the 300 ha unfortified Islamic city of **Velyki Kuchugury** (47,68 N, 35,20 E, with a Friday mosque, glazed minaret, subterranean-heated *hammam*, water conduits and **5,5k - 6,0k** inhabitants), paired with **Mamai-Surka** (1 100+ Golden Horde burials), **Konski Vody / Mechetna Mohyla** (**4,0k**) and **Oleshye** (`2,0k`), while Italian charts mapped **Porto Pisano** (**Belosaraiskaya**, `1,5k`) on the Azov coast of Sub-region V [20][21].

The Black Death, Blue Waters, Timur's 1395AD Campaign and the 1400AD Terminal Benchmark

Between 1346AD and 1400AD, the interior steppes of Sub-regions II, III, IV and V suffered severe demographic contraction driven by the **Black Death** (`1346 – 1348AD`), the **Great Troubles** of the Golden Horde (`1359 – 1380AD`), Grand Duke Algirdas's victory at the **Battle of Blue Waters** (`1362AD`, on the Syniukha River in Sub-region III), **Timur's 1395AD campaign** (which systematically razed Kuchugury, Mamai-Surka, Konski Vody and the Donets settlements) and the **Battle of the Vorskla River** (`1399AD`) [21][49]. By contrast, **Sub-region I (Budjak)** and the coastal ports of **Sub-region II** exhibited much greater resilience as the **Principality of Moldavia** (under Petru I Musat and Alexander the Good, `1390s – 1400AD`) fortified **Cetatea Alba / Aq Qerman** (`12,0k` in 1400AD), **Kiliia / Chilia** (`6,0k`) and **Smil / Izmail** (`3,5k`),

while Grand Duke Vytautas of Lithuania established fortified customs ports at **Dashov / Kara-Kerman** (Ochakiv, 2,8k in 1400AD), **Khadzhibey / Koczubij** (Odesa Bay, 2,5k), **Sokolets**, **Balyklei** and **Tyagin** [11].

At the **1400AD terminal benchmark**, the total population of the Wild Fields stood at **155,0k** (24,0k in **Sub-region I: Budjak** [15,5% of the regional total on 5,5% of the land area], 34,0k in Sub-region II, 34,0k in Sub-region III, 35,0k in Sub-region IV and 28,0k in Sub-region V).

Quantitative Synthesis, Visual Analytics and Complete Benchmark Series (10000BC - 1400AD)

Figures 2, 3 and 4 and Table 2 present the complete demographic synthesis for the Wild Fields across all 25 chronological benchmarks from 10000BC to 1400AD across the five natural sub-regions.

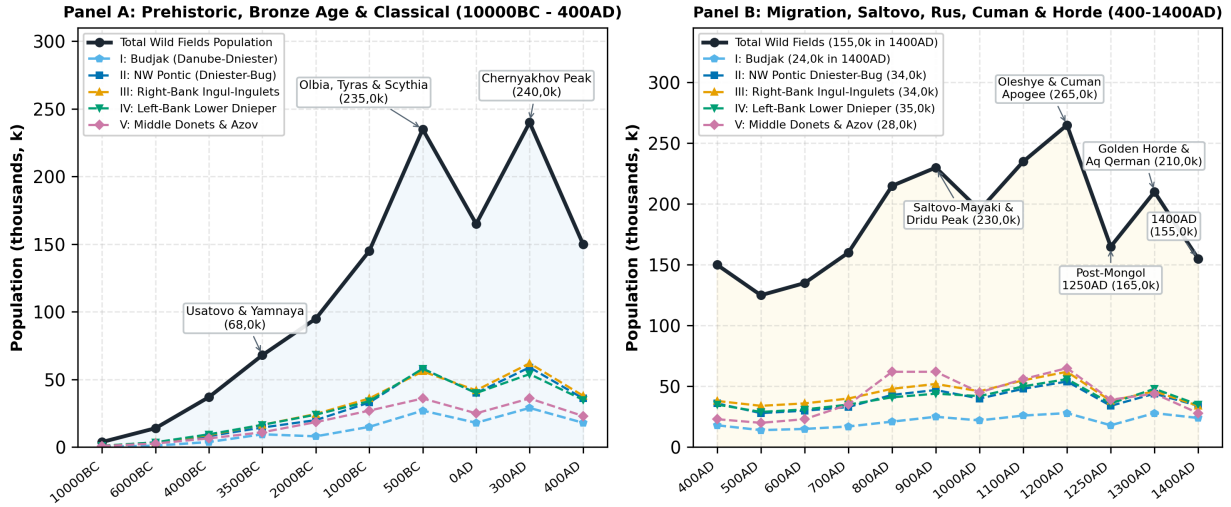


Figure 2. Long-Run Demographic Trajectory of the Wild Fields (238 000 km2) across Five Natural Sub-Regions Including Budjak: Prehistoric, Bronze Age & Classical Cycles (10000BC - 400AD, Panel A) and Migration, Saltovo-Mayaki, Kievan Rus, Cuman & Golden Horde Horizons (400AD - 1400AD, Panel B).

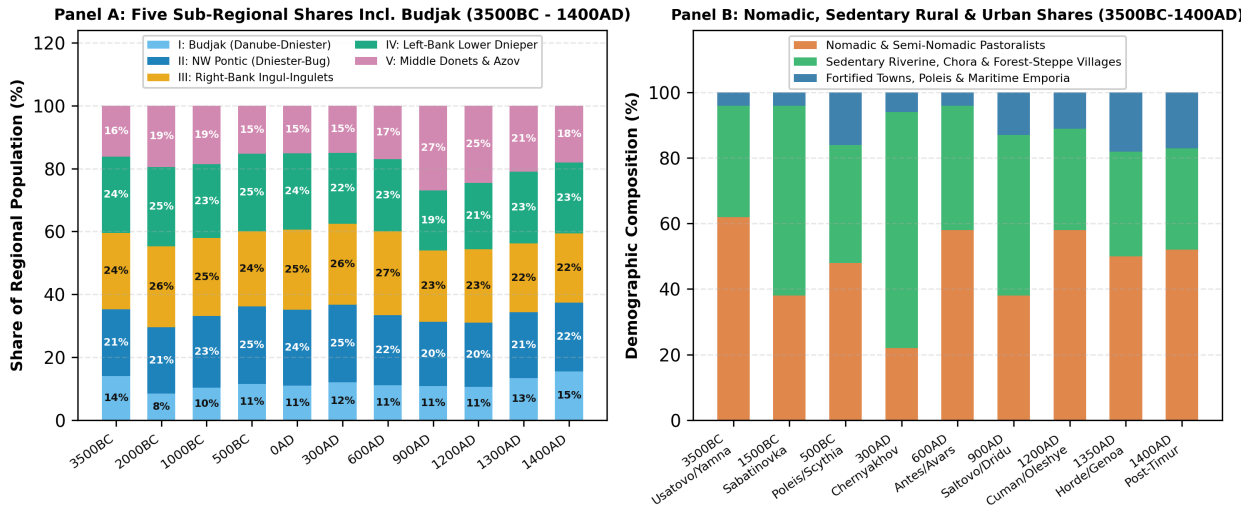


Figure 3. Sub-Regional Population Shares across Five Natural Sub-Regions I-V Including Budjak (3500BC - 1400AD, Panel A) and Structural Evolution of Nomadic-Pastoral, Sedentary Rural and Urban Shares (Panel B).

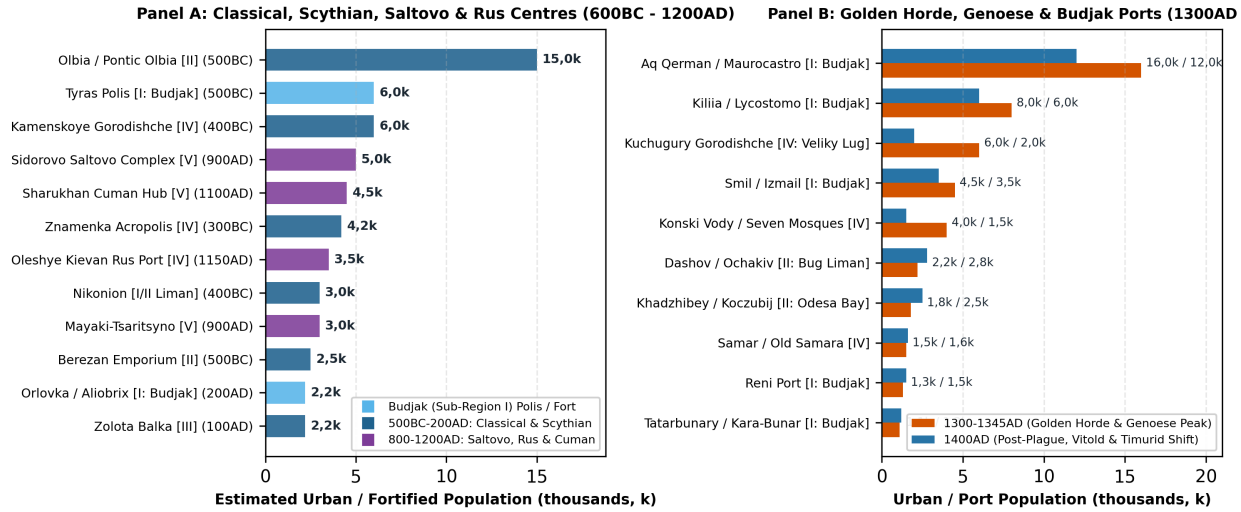


Figure 4. Comparative Urban and Fortified Settlement Hierarchies in the Wild Fields across Classical, Scythian, Saltovo & Kievan Rus Horizons (600BC - 1200AD, Panel A) and Golden Horde, Genoese & Moldavian-Budjak Centres (1300AD vs 1400AD, Panel B).

Date	I: Budjak	II: NW Pontic	III: R-Bank	IV: L-Bank	V: Donets	Total	Primary Archaeological / Historical Regime & Anchor Sources
10000BC	0,30k	0,80k	0,95k	0,95k	0,80k	3,8k	Epigravettian bison & horse hunters (Anetovka II, Amvrosievka)
8000BC	0,55k	1,45k	1,7k	1,7k	1,4k	6,8k	Kukrek & Grebeniki Mesolithic camps & rapids cemeteries (Mirnoe [Budjak], Vasylivka)
6000BC	1,1k	3,1k	3,5k	3,6k	2,7k	14,0k	Bug-Dniester, Sursko-Dnieper & Danube-Lakes Early Neolithic villages
5000BC	1,9k	5,3k	6,0k	6,2k	4,6k	24,0k	Dnieper-Donets II & Mariupol Neolithic collective cemeteries (Vovnigi, Mariupol)
4000BC	3,8k	8,2k	9,0k	9,5k	6,5k	37,0k	Bolgrad-Aldeni (Gumelnita) in Budjak; Sredny Stog & Dereivka horse-keepers
3500BC	9,5k	14,5k	16,5k	16,5k	11,0k	68,0k	Budjak Kurgan & Usatovo maritime chiefdoms; Early Mikhailovka Eneolithic peak
3000BC	8,5k	14,5k	19,0k	19,0k	14,0k	75,0k	Yamnaya (Pit-Grave) wagon pastoralism (Nerushai, Mikhailovka stone citadel)
2000BC	8,0k	20,0k	24,5k	24,0k	18,5k	95,0k	Catacomb (Ingul/Donets) & Babino Middle Bronze Age pastoralism

(Continued from previous page)

Date	I: Budjak	II: NW Pontic	III: R-Bank	IV: L-Bank	V: Donets	Total	Primary Archaeological / Historical Regime & Anchor Sources
1000BC	15,0k	33,0k	36,0k	34,0k	27,0k	145,0k	Sabatinovka-Coslogeni (Budjak), Srubnaya & Belozerka Late Bronze village boom
500BC	27,0k	58,0k	56,0k	58,0k	36,0k	235,0k	Classical Greek Poleis (Tyras [Budjak], Olbia, Nikonion) & Royal Scythia (Kamenskoye)
0AD	18,0k	40,0k	42,0k	40,0k	25,0k	165,0k	Sarmatian migrations & early Late Scythian Lower Dnieper stone fortresses
200AD	24,0k	50,0k	52,0k	48,0k	31,0k	205,0k	Roman-allied Tyras, Aliobrix [Budjak] & Olbia; 16 Late Scythian fortresses
300AD	29,0k	59,0k	62,0k	54,0k	36,0k	240,0k	Chernyakhov sedentary agro-pastoral peak across Budjak, Liman & Dnieper valleys
400AD	18,0k	36,0k	38,0k	35,0k	23,0k	150,0k	Post-375AD Hunnic collapse of Chernyakhov villages & coastal poleis
500AD	14,0k	28,0k	34,0k	29,0k	20,0k	125,0k	Early Medieval Kutrigur/ Utigur nomadism; Antae/ Penkovka at Dnieper Rapids
600AD	15,0k	30,0k	36,0k	31,0k	23,0k	135,0k	Avar & Old Great Bulgaria steppe transit; Danubian & Rapids refugia
700AD	17,0k	33,0k	40,0k	35,0k	35,0k	160,0k	Asparukh's Onglos in Budjak; Voznesenka princely camp; early Khazar rule
800AD	21,0k	43,0k	48,0k	41,0k	62,0k	215,0k	Dridu (First Bulgarian Empire) villages in Budjak; Saltovo-Mayaki boom on Donets
900AD	25,0k	47,0k	52,0k	44,0k	62,0k	230,0k	Saltovo-Mayaki (Sidorovo 5k) & Budjak Dridu apogee; Ulichi & Tivertsi towns
1000AD	22,0k	40,0k	46,0k	42,0k	45,0k	195,0k	Post-965AD Saltovo collapse; Pecheneg steppe dominance; Kievan Rus Oleshye
1100AD	26,0k	48,0k	55,0k	50,0k	56,0k	235,0k	Polovtsian winter capitals (Sharukhan) & Rus Oleshye, Protolche & Berladniki

(Continued from previous page)

Date	I: Budjak	II: NW Pontic	III: R-Bank	IV: L-Bank	V: Donets	Total	Primary Archaeological / Historical Regime & Anchor Sources
1200AD	28,0k	54,0k	62,0k	56,0k	65,0k	265,0k	High Medieval Cuman-Rus commercial & agro-pastoral apogee (Maurocastro, Oleshye)
1240AD	26,0k	50,0k	57,0k	52,0k	60,0k	245,0k	Pre-Mongol eve (subsequent to 1223AD Battle of Kalka River)
1250AD	18,0k	34,0k	38,0k	36,0k	39,0k	165,0k	Post-1240AD Mongol conquest demographic trough across interior watersheds
1300AD	28,0k	44,0k	46,0k	48,0k	44,0k	210,0k	Golden Horde & Genoese urban revival (Aq Qerman 16k & Kiliia 8k in Budjak; Kuchugury 6k)
1400AD	24,0k	34,0k	34,0k	35,0k	28,0k	155,0k	Post-Plague & 1395AD Timurid steppe trough; Moldavian-Genoese Budjak & Lithuanian ports

Table 2. Complete Sub-Regional and Total Demographic Benchmark Series for the Wild Fields (238 000 km2) across Five Natural Sub-Regions Including Budjak, 10000BC - 1400AD.

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